Discovering Cooperative Pipelines: Autoresearch for Sequential Social Dilemmas

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Abstract

We study *two-level autoresearch* for cooperation: an outer-loop AI agent autonomously redesigns the inner-loop pipeline of an LLM policy-synthesis system for multi-agent Sequential Social Dilemmas (SSDs). A *researcher agent* $\mathcal{R}$ (run as a coding agent) reads the inner-loop source code, edits system prompts, feedback functions, helper libraries, and iteration logic, runs evaluations, and decides what to keep, following the *autoresearch* paradigm. Across two games (Cleanup and Gathering), two policy-synthesizer LLMs, and two welfare objectives (utilitarian efficiency and Rawlsian maximin), the researcher reliably exceeds hand-designed baselines, sharply tightens run-to-run variance, and outperforms prompt-only optimization. The discovered pipelines are objective-dependent: under maximin the researcher independently rediscovers *duty rotation* as a fairness mechanism, supporting an *information-design* reading in which the researcher chooses what to reveal to the boundedly rational synthesizer. Code at `https://github.com/anon590/llm-policies-social-dilemmas`.

1 Introduction

Sequential Social Dilemmas (SSDs) [1] are the multi-agent analogue of the prisoner's dilemma in temporally rich Markov games: individually rational play leads to collectively suboptimal outcomes through pollution, over-harvesting, or open conflict. Standard multi-agent reinforcement learning (MARL) struggles in this regime due to credit assignment, non-stationarity, and large joint action spaces [2]. A complementary approach, recently introduced by Gallego [3], replaces parameter-space optimization with *algorithm-space synthesis*: a frozen LLM writes a Python policy function, evaluates it in self-play, and iteratively refines it from performance feedback. A single generation step can produce coordination logic (territory partitioning, role assignment, conditional cooperation) at a sample efficiency several orders of magnitude beyond what gradient-based MARL achieves on the same environments.

This shifts where the design problem lives, rather than removing it. The inner-loop pipeline that drives the synthesizer has many free parameters: which system prompt, which feedback variables, which helper functions, how many refinement steps. Each materially affects the resulting policies, and prior work tuned them by hand. A natural question follows: *can an AI agent design the pipeline?*

We answer affirmatively with a two-level autoresearch framework. An *outer-loop researcher agent* (Claude Opus 4.6, run as a coding agent) edits the source files of an *inner-loop policy synthesizer* (another LLM), runs evaluations on held-out seeds, and keeps modifications that improve a fixed welfare objective Φ. The outer agent operates on an ordinary git repository (reading code, writing diffs, running shell commands, etc) without task-specific scaffolding, mirroring the autoresearch paradigm of Karpathy [4] for single-GPU LLM pretraining. Although the inner-loop SSDs are gridworld benchmarks rather than physical systems, the outer-loop discovery process itself runs under conditions a deployed discovery agent faces: noisy multi-seed evaluations, stochastic code generation, an LLM-evaluation budget that bounds how often Φ can be queried, and a heterogeneous code repository the agent must navigate end-to-end on its own.

Our contributions are: i) a general two-level framework that delegates the design of an LLM synthesis pipeline to a coding agent operating on a real software repository (Section 3); ii) the first instantiation of the autoresearch paradigm in a multi-agent decision-making domain, with experiments across two SSDs (Cleanup and Gathering), two policy LLMs, and two welfare objectives (utilitarian efficiency U and Rawlsian maximin $\min_i R_i$) (Section 4); and iii) a mechanism-design interpretation supported by the qualitatively different pipelines the agent produces under different welfare objectives, including the autonomous discovery of *time-based duty rotation* as a fairness mechanism, never found under efficiency optimization.

2 Background

We build on the iterative LLM policy synthesis framework of Gallego [3], which serves as the (frozen) inner loop of our two-level system. This section recalls the SSD formalism, the social outcome metrics, and the base synthesis loop.

2.1 Sequential Social Dilemmas

A Sequential Social Dilemma is a partially observable Markov game $\mathcal{G} = \langle N, \mathcal{S}, \{\mathcal{A}_i\}_{i=1}^N, T, \{R_i\}_{i=1}^N, H\rangle$ with N agents, state space $\mathcal{S}$ (the gridworld configuration), per-agent action spaces $\mathcal{A}_i$, transition function T, reward functions R_i, and episode horizon H [1]. Beyond the dilemma's matrix-game structure, SSDs add temporal richness: agents must learn *when* and *where* to cooperate, not just *whether* to.

We study two canonical SSDs that capture complementary dilemma types.

Cleanup [5] is a *public goods provision* game (N=10). The map has two regions: a river that accumulates waste, and an orchard where apples grow. Apples regrow only when river pollution is below threshold. Each agent can fire a cleaning beam (cost -1) that removes waste, collect apples ($+1$ each), or fire a tagging beam (cost -1, inflicting -50 on the target and removing it for 25 steps). The dilemma: cleaning is costly but its benefits are public, so purely self-interested agents free-ride.

Gathering [1, 6] is a *common pool resource* game (N=4). Agents collect shared apples on a fixed respawn timer and may fire tagging beams to temporarily remove rivals. The dilemma: agents can coexist and share resources, or aggress to monopolize them; aggression wastes time and reduces total welfare.

Both games use 8–9 discrete actions (movement, rotation, beam, stand, optionally clean) and episodes of H=1000 steps. The two dilemmas differ in cost structure: asymmetric provision (cleaners pay, all benefit) vs. symmetric restraint (every agent faces the same temptation), a distinction that drives our experimental findings.

Social metrics. Following Perolat et al. [6], let $R_i = \sum_{t=0}^{H-1} r_i^t$ denote agent i's episode return. We evaluate four social outcomes:

$$U = \frac{1}{H}\sum_{i=1}^{N} R_i \qquad \text{(efficiency)} \qquad (1)$$

$$E = 1 - \frac{\sum_{i,j} |R_i - R_j|}{2N\sum_i R_i} \qquad \text{(equality)} \qquad (2)$$

$$S = \frac{1}{N}\sum_{i=1}^{N} \bar{t}_i \qquad \text{(sustainability)} \qquad (3)$$

$$P = \frac{1}{H}\sum_{t=0}^{H-1} \left|\{i : \text{active}_i^t\}\right| \qquad \text{(peace)} \qquad (4)$$

where $\bar{t}_i$ is the mean timestep at which agent i collects positive reward (higher $\Rightarrow$ resources preserved later) and active_i^t indicates that agent i is not tagged out at step t. We additionally consider the *maximin* (Rawlsian) welfare criterion $\min_i R_i$, which optimizes the worst-off agent's return and serves as the second objective in our experiments.

2.2 Iterative LLM Policy Synthesis

Let Π denote the space of *code-based policies*: deterministic functions $\pi : \mathcal{S} \times [N] \to \mathcal{A}$ expressed as executable Python code. Each policy has access to the full environment state and a library of helpers (BFS pathfinding, beam targeting, coordinate transforms). This state access is a deliberate design choice: programmatic policies operate in algorithm space rather than the reactive observation-to-action space of neural policies, which lets a single LLM generation step encode rich coordination logic.

A frozen LLM $\mathcal{M}$ acts as the *policy synthesizer*. Given a system prompt p describing the environment API and a feedback prompt q_k, it produces a new policy

$$\pi_{k+1} = \mathcal{M}(p,\ q(\pi_k, \mathcal{F}_k))\,,$$

where π_k is the previous policy (its source code) and $\mathcal{F}_k$ is the evaluation feedback. All N agents execute the same program π_k in self-play. We stress that this is *symmetric*, not *behaviorally homogeneous*: since π_k takes `agent_id` as an argument, a single shared program can induce distinct per-agent behaviors (cleaner vs. gatherer assignment, time-rotated duty cycles, partitioned territories; see the synthesized policies in Appendix E). What is shared is the source code; the sampled action distributions can differ across agents. Evaluation over a set of random seeds S yields the mean per-agent return $\bar{r}_k$ and the social metrics vector $\mathbf{m}_k = (U_k, E_k, S_k, P_k)$. Each generated policy passes an AST-based safety check (blocking `eval`, file I/O, network access) followed by a short smoke test; failures trigger regeneration (up to R attempts) with the error message appended to the prompt.

Feedback. We package the previous policy's source code together with all available evaluation signals:

$$\mathcal{F}_k = \big(\text{code}(\pi_k),\ \bar{r}_k,\ \mathbf{m}_k,\ \mathbf{d}\big), \tag{5}$$

where $\mathbf{d}$ contains natural-language definitions of each social metric. The LLM consumes $\mathcal{F}_k$ to revise and improve the policy. This is a starting point; the choice of feedback content is a single design decision within a much larger pipeline configuration space: our two-level framework (Section 3) opens the full space to automated search.

3 Two-Level Framework

We introduce a two-level system where a *researcher agent* $\mathcal{R}$ autonomously discovers configurations that optimize the output of an inner-loop system. While we instantiate this for multi-agent policy synthesis, the architecture is general: any pipeline where an LLM generates artifacts, evaluates them, and iterates can serve as the inner loop. The fundamental insight is that the entire inner-loop codebase is a designable artifact that a code-based agent can search over. Figure 1 illustrates the architecture.

3.1 Configuration Space

Let $\mathcal{C}$ denote the space of *pipeline configurations*. Each configuration $c \in \mathcal{C}$ specifies the full inner-loop setup:

$$c = (p,\ \phi,\ \mathcal{H},\ \iota) \tag{6}$$

where p is the system prompt, ϕ is the feedback construction function (which metrics and diagnostics to include, how to frame it, whether to inject adaptive hints and thresholds, etc), $\mathcal{H}$ is the helper function library (auxiliary functions for pathfinding, getting aggregates of useful environemnt quantities, etc.), and ι specifies the iteration logic (number of inner iterations K, sampling strategy). Table 1 provides concrete examples. The validation pipeline is part of the frozen inner-loop infrastructure rather than a configurable component, the researcher cannot modify it in our experiments to prevent reward hacking.

The hand-designed feedback of [3] corresponds to a single fixed instantiation of ϕ. Our framework opens the full configuration space to automated search.

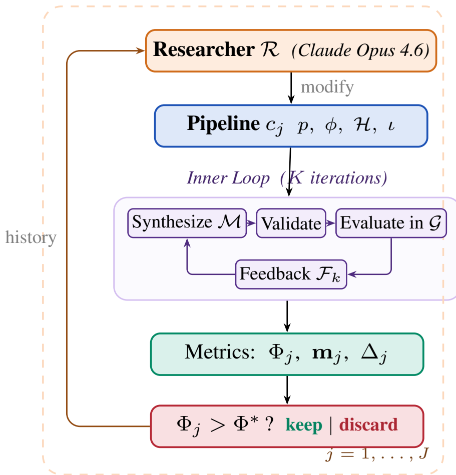

Figure 1: Two-level automated research framework (Algorithm 1).

Algorithm 1 Two-Level Automated Research

Require: Game $\mathcal{G}$, policy synthesizer $\mathcal{M}$, researcher $\mathcal{R}$, system prompt $p_{\mathcal{R}}$, initial config c_0, outer iterations J, ground-truth objective Φ, held-out seeds S_{ho}

Ensure: Best configuration c^*, best policy π^*

$\pi_0^* \leftarrow \textsc{InnerLoop}(\mathcal{M}, \mathcal{G}, c_0)$

$\Phi_0 \leftarrow \textsc{Eval}(\pi_0^*;\ \mathcal{G},\ S_{\text{ho}})$

history $\leftarrow \{(c_0, \Phi_0, \mathbf{m}_0, \varnothing)\}$

for $j = 1, \ldots, J$ **do**

  $c_j \leftarrow \mathcal{R}(p_{\mathcal{R}},\ \textsc{code}(c_{j-1}),\ \textit{history})$

  if $\neg\,\textsc{ValidateConfig}(c_j)$ **then** retry $\leq R$

  $\pi_j^* \leftarrow \textsc{InnerLoop}(\mathcal{M}, \mathcal{G}, c_j)$

  $\Phi_j \leftarrow \textsc{Eval}(\pi_j^*;\ \mathcal{G},\ S_{\text{ho}})$

  $\Delta_j \leftarrow \textsc{Diff}(c_{j-1}, c_j)$ *// code diff*

  history $\leftarrow$ *history* $\cup \{(c_j, \Phi_j, \mathbf{m}_j, \Delta_j)\}$

end for

$j^* \leftarrow \arg\max_j \Phi_j$

return $c_{j^*},\ \pi_{j^*}^*$

Table 1: Configuration components modifiable by the researcher, with concrete edits that $\mathcal{R}$ made in our runs. The environment simulator, ground-truth evaluation, and policy LLM weights are frozen.

	Scope	Concrete edits by $\mathcal{R}$ (see appendix)
p	System prompt	Multi-section strategic briefing with explicit duty-rotation template `(agent_id + step//T) % n` (App. D.2, Listing 3)
ϕ	Feedback fn	Thresholded diagnostics: "FAIRNESS ALERT" fires when $\min_i R_i{<}0$; "DO NOT REGRESS" guard fires when $U{\geq}2.5$ (App. D.3, Listing 4)
$\mathcal{H}$	Helper library	BFS-Voronoi territories with respawn-timer awareness; band-based apple zoning by `agent_id` (App. D.1, Listings 2, 1)
ι	Iteration logic	Per-condition $K{\in}\{2,3\}$; $\lvert S\rvert$ walked 5→8→12 for variance control; thinking budget 10–32k (App. D.4, Table 6)

3.2 Inner Loop (Policy Synthesis)

Given a configuration c, the inner loop executes K iterations of LLM policy synthesis:

$$\pi_c^* = \textsc{InnerLoop}(\mathcal{M}, \mathcal{G}, c) \tag{7}$$

Each iteration k proceeds in four stages, following [3]:

1. **Synthesize.** The policy LLM $\mathcal{M}$ receives the system prompt p, the previous policy's source code π_{k-1}, and feedback $\mathcal{F}_{k-1}$ constructed by ϕ. It generates a new Python policy function π_k that has access to full environment state and the helper library $\mathcal{H}$.
2. **Validate.** The generated code undergoes AST-based safety checks (blocking dangerous operations such as file I/O and network access) followed by a short smoke test. Failures trigger re-generation (up to R retries), with the error message appended to the prompt.
3. **Evaluate.** All N agents execute the same policy π_k in self-play over $|S|$ random seeds (note the policy is conditional on `agent_id`). The evaluation yields the mean per-agent reward $\bar{r}_k$ and the social metrics vector $\mathbf{m}_k = (U_k, E_k, S_k, P_k)$.
4. **Feedback.** The feedback function ϕ constructs the prompt for the next iteration from $(\pi_k, \bar{r}_k, \mathbf{m}_k)$, packaging the previous policy's source code together with the scalar reward, the social metrics vector, their natural-language definitions, and any adaptive diagnostics that ϕ injects.

The inner loop output is evaluated on held-out seeds via a fixed ground-truth objective:

$$\Phi(c) = \textsc{Eval}(\pi_c^*;\ \mathcal{G},\ S_{\text{held-out}}) \tag{8}$$

where Φ maps from social outcomes to a scalar. We consider two alternative objectives to maximize:

$$\Phi_U = U = \frac{1}{H}\sum_{i=1}^{N} R_i \qquad \text{(utilitarian efficiency)} \tag{9}$$

$$\Phi_{\min} = \min_i R_i \qquad \text{(Rawlsian maximin)} \tag{10}$$

Φ_U rewards collective throughput and is indifferent to how reward is distributed across agents, whereas$\Phi_{\min}$ instead optimizes for the worst-off agent, pressuring the researcher toward configurations that distribute the cost of cooperation.

3.3 Outer Loop (Automated Research)

The researcher agent $\mathcal{R}$ iteratively modifies the pipeline configuration. Following the autoresearch paradigm [4], $\mathcal{R}$ operates on the inner-loop codebase as a modifiable artifact, proposing changes, observing outcomes, and refining. The procedure is formalized in Algorithm 1.

At each outer iteration j, the researcher $\mathcal{R}$ receives: i) the **full source code** of the current configuration c_{j-1} (prompts, feedback construction, helpers, iteration logic); ii) the **experiment history**: for each prior iteration, the code diff Δ_i, ground-truth score Φ_i, and social metrics vector $\mathbf{m}_i$; iii) the **environment source code** (read-only), enabling the researcher to reason about game mechanics. The researcher proposes a new configuration c_j by generating code modifications. Concretely, $\mathcal{R}$ is a coding agent (Claude Code CLI) that operates on a real software repository: it reads and edits Python source files, runs shell commands, inspects evaluation outputs, and commits changes to a dedicated git branch, following the same workflow a human researcher would follow.

3.4 Connection to Automated Mechanism Design

The two-level structure admits a mechanism design interpretation. The researcher $\mathcal{R}$ acts as a *mechanism designer*: it controls the information structure (what metrics to reveal, how to frame them), the action space (what helper functions are available), and the incentive structure (how feedback is presented) under which the policy synthesizer $\mathcal{M}$ operates. The synthesizer acts as the *agent* within the designed mechanism.

This connects to the automated mechanism design literature [7], where a principal designs rules to induce desired behavior from self-interested agents. In our setting:

- The **principal** is the researcher $\mathcal{R}$, optimizing the ground-truth objective Φ.
- The **agent** is the synthesizer $\mathcal{M}$, optimizing per-agent reward as instructed.
- The **mechanism** is the configuration c: prompts, feedback, helpers, iteration logic.
- The **outcome** is the social welfare of the resulting multi-agent policy π_c^*.

A crucial distinction from classical mechanism design: $\mathcal{M}$ follows instructions but has *bounded rationality* in the sense that its ability to synthesize effective policies depends on the information and tools provided. The researcher's task is thus closer to *information design* [8]: choosing what to reveal to help $\mathcal{M}$ navigate the cooperation–defection tension. Our experiments (Section 4) show that the researcher designs qualitatively different information structures depending on the welfare objective Φ, supporting this interpretation empirically.

4 Experiments

4.1 Setup

We conduct 12 autonomous researcher runs across a factorial design. The researcher agent $\mathcal{R}$ is Claude Opus 4.6, invoked via the Claude Code CLI as a coding agent. Each run operates on a dedicated git branch of a real Python codebase: $\mathcal{R}$ edits source files in `pipeline/`, executes evaluation scripts, reads metric outputs, and iterates, without human intervention.

Design. For Cleanup: 2 policy LLMs $\times$ 2 objectives $\times$ 2 replications $=$ 8 runs. For Gathering: 2 policy LLMs $\times$ 1 objective $\times$ 2 replications $=$ 4 runs. Maximin runs are unnecessary for Gathering because efficiency optimization alone achieves close to perfect equality.

Table 2: Results. Mean / max across the runs per condition. **Bold** marks the best mean per metric within each Policy LLM × Game sub-block. Baseline: unmodified, hand-designed pipeline from [3]. GEPA: automated prompt optimization [9] with matched compute budget.

Policy LLM	Target	U	E	$\min_i R_i$
Cleanup — Baseline				
Gemini 3.1 Pro	—	1.93/2.70	0.17/0.62	−159/−84
Sonnet 4.6	—	0.86/1.56	−0.02/0.25	−151/−59
Cleanup — GEPA [9]				
Gemini 3.1 Pro	Φ_U	1.34/1.37	−0.10/−0.05	−149/−126
	$\Phi_{\min}$	1.76/2.76	0.90/0.96	143/245
Sonnet 4.6	Φ_U	1.04/1.20	−0.59/−0.21	−164/−126
	$\Phi_{\min}$	−0.63/1.60	0.77/1.00	−147/6
Cleanup — Two-level Autoresearch				
Gemini 3.1 Pro	Φ_U	**3.20**/3.25	0.55/0.61	−211/−182
	$\Phi_{\min}$	3.16/3.19	**0.98**/0.98	**290**/296
Sonnet 4.6	Φ_U	**3.12**/3.14	0.66/0.70	−196/−146
	$\Phi_{\min}$	2.57/2.93	**0.91**/0.97	**179**/200
Gathering — Baseline				
Gemini 3.1 Pro	—	2.04/2.42	0.90/0.98	412/571
Sonnet 4.6	—	0.03/0.03	0.54/0.54	0/0
Gathering — GEPA [9]				
Gemini 3.1 Pro	Φ_U	2.08/2.35	0.94/0.96	436/518
Sonnet 4.6	Φ_U	1.20/1.23	0.63/0.71	86/123
Gathering — Two-level Autoresearch				
Gemini 3.1 Pro	Φ_U	**2.49**/2.51	**0.98**/0.98	**582**/582
Sonnet 4.6	Φ_U	**2.52**/2.52	**0.96**/0.98	**576**/597

Models. Policy synthesizer $\mathcal{M}$: Gemini 3.1 Pro (Google) or Claude Sonnet 4.6 (Anthropic), to recent state-of-the-art LLMs. Both use extended thinking. We additionally test with Gemma 4 26B-A4B-IT (Google), a smaller open-weight model, to probe the framework's behavior when the policy synthesizer has substantially lower capability (Appendix C).

Baselines. The hand-designed feedback configuration from prior work [3] serves as the initial pipeline c_0 for all runs. On Cleanup ($N{=}10$), this baseline achieves $U{=}1.93$/2.70 (mean/max) with Gemini and $U{=}0.86$/1.56 with Sonnet. We additionally compare against GEPA [9], an automated prompt optimization method that iteratively refines the system prompt via LLM reflection. GEPA optimizes only the system prompt p, whereas our researcher modifies the full pipeline $(p, \phi, \mathcal{H}, \iota)$. We give GEPA a matched compute budget: the same number of optimization steps as outer iterations used by our method, so all in all both methods use a comparable number of environment evaluations.

4.2 Results

Table 2 presents the main results, comparing our autoresearch framework against the hand-designed baseline of [3] and GEPA [9].

Finding 1: The researcher reliably improves over hand-designed baselines and outperforms prompt-only optimization. Every run improves substantially regardless of starting point (Figure 2). On Cleanup, autoresearch lifts both LLMs to $U \approx 3.1$–3.2 from baselines of 1.93 (Gemini) and 0.86 (Sonnet), nearly closing the gap between them; on Gathering, all four runs converge to $U \in [2.47, 2.52]$ from baselines spanning 0.03–2.42. Run-to-run spread is tight (gaps within 0.05 on Cleanup-Φ_U), suggesting the researcher reliably finds the performance ceiling of each policy LLM via the pipeline modifications it discovers (helpers, prompts, and feedback; see appendices D.1–D.3 for a selection of them). At matched environment queries, autoresearch beats GEPA by 2–3× on Cleanup (both Φ_U and $\Phi_{\min}$) and 20% on Gathering, with the gap widening for the weaker policy LLM: GEPA-Sonnet under $\Phi_{\min}$ can collapse to a pathological "everyone cleans, nobody

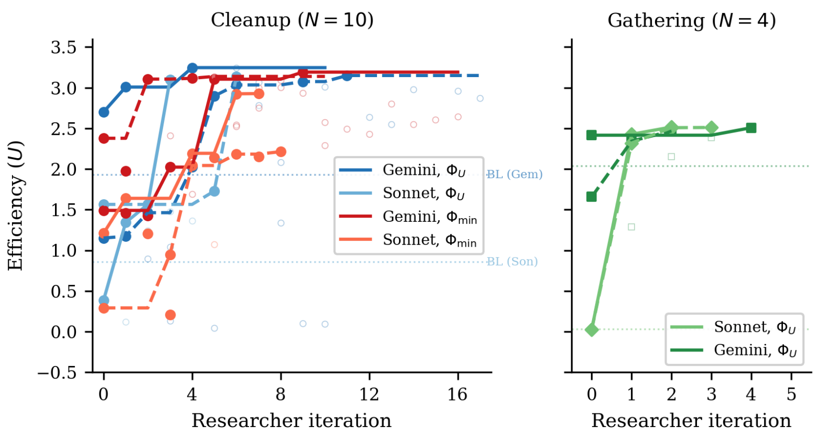

Figure 2: Efficiency (U) across researcher iterations for all 12 runs. *Left*: Cleanup with 8 runs across 2 LLMs $\times$ 2 objectives (solid lines = kept experiments, open circles = discarded). Dashed horizontal line: hand-designed baseline from [3]. All runs converge to $U \approx 3.1$–3.2 despite diverse starting points. *Right*: Gathering with 4 runs (2 per LLM). All converge to $U \approx 2.5$ within 2–4 iterations despite baselines ranging from 0.03 to 2.42.

eats" regime (U=-2.87, E=1.00), while autoresearch-Sonnet reaches $\min_i R_i \approx 200$ reliably. Modifying the full pipeline, not just the prompt, is what closes these gaps.

Finding 2: No efficiency–fairness tradeoff in Cleanup (Gemini). Maximin-optimized Gemini pipelines sacrifice only 1% efficiency (U: 3.16 vs. 3.20) while achieving near-perfect equality (E: 0.98 vs. 0.55) and transforming maximin from deeply negative baselines (-99 to -84) to $\min_i R_i = 290$. The researcher discovers *fair duty rotation* (Listing 6; primed by the prompt rewrite of Listing 3)—time-based cycling using `agent_id` and `env._step_count`—simultaneously improving worst-off welfare *and* collective output. Because cleaning is a public good, distributing the cleaning cost fairly ensures enough cleaners to sustain apple production.

For Sonnet, there is a moderate tradeoff: efficiency drops from 3.12 to 2.57 under maximin optimization. The gap reflects Sonnet's harder time implementing complex coordination mechanisms (role rotation, zone assignment) from strategic hints alone.

Finding 3: Game structure determines whether fairness requires explicit optimization. In Cleanup, where cleaning costs are borne asymmetrically (cleaners pay -1, free-riders collect apples), baseline equality ranges from E=0.04 to 0.62, and maximin optimization is required to reach $E > 0.9$. In Gathering, where all agents face a symmetric landscape, efficiency optimization alone achieves $E > 0.94$ across all 4 runs: no separate maximin runs are needed. This generalizes: in *provision* dilemmas with asymmetric costs, fairness requires designed mechanisms (role rotation, duty sharing); in *restraint* dilemmas with symmetric costs, fairness emerges as a free byproduct of efficient coordination. The researcher independently discovers this, it creates role differentiation pipelines only for Cleanup, and pure spatial-coordination pipelines for Gathering.

Finding 4: Convergent discovery of qualitatively different strategies per objective. Despite fully independent runs, the researcher converges on the same core strategies within each condition (Table 3, Appendix A). In Cleanup, waste-counting helpers and spatial zone partitioning appear across all runs. The main qualitative difference is *role rotation*, appearing in 4/4 maximin runs (Listings 6, 7) but 0/4 efficiency runs (Listing 5). Under efficiency optimization, the researcher assigns static cleaning roles (some agents always clean), achieving high collective output at the cost of equality. Under maximin optimization, it discovers rotation as a fairness mechanism. In Gathering, the researcher discovers BFS-Voronoi territory partitioning and respawn-timer awareness (Listings 2, 8), with no role differentiation. Thus, optimization is purely spatial (who collects which apples) and temporal (respawn-aware positioning).

Common failure modes. Several patterns led to discarded autoresearch iterations across all runs: (1) *over-prescription*: adding too many strategic hints confuses the policy LLM, causing generation failures; (2) *iteration regression*: $K{=}4$ or $K{=}5$ inner iterations often cause catastrophic regression as the LLM "over-refines" a working policy; (3) *feedback overload*: verbose per-agent diagnostics cause over-correction rather than gradual improvement.

5 Related Work

Sequential social dilemmas. SSDs were introduced by Leibo et al. [1] (Gathering) and extended to public-goods settings by Hughes et al. [5] (Cleanup); Perolat et al. [6] formalized the social outcome metrics (U, E, S, P) used here. Subsequent work studies inequity aversion, intrinsic motivation, and reputational mechanisms for promoting cooperation in MARL agents. We complement this line by automating the search for cooperative *programs* rather than evolving neural policies, and by treating the welfare objective Φ as a designable parameter that the outer agent optimizes for, obtaining qualitatively different cooperative behaviors (static role assignment vs. duty rotation) under different objectives without changing the environment.

LLMs for policy and program synthesis. FunSearch [10] evolves programs for combinatorial discovery; Eureka [11] synthesizes reward functions from environment source code; Voyager [12] and Code as Policies [13] generate executable skill code for single-agent embodied control; ReEvo [14] evolves heuristics with reflective feedback. These works target single-agent settings or non-strategic optimization. Gallego [3], on which our inner loop is based, extended LLM program synthesis to the multi-agent SSD setting, where one program must coordinate N self-play copies. Our contribution is one level above: rather than tuning the synthesizer for one task, we let an outer agent rewrite the synthesis pipeline itself.

LLM reflection and prompt optimization. Reflexion [15], Self-Refine [16], OPRO [17], and GEPA [9] demonstrate that structured verbal feedback loops improve LLM outputs; ERL [18] internalizes such reflection via self-distillation. These methods optimize the prompt or the model's reasoning trajectory in isolation. Our framework instead modifies the entire surrounding pipeline (system prompt, feedback construction, helper library, iteration logic) making prompt optimization one component of a strictly larger search space. The empirical gap is large: Section 4 shows full-pipeline autoresearch achieving $3\times$ higher efficiency in Cleanup and 37% higher in Gathering than GEPA at matched compute, with run-to-run spread an order of magnitude tighter on the harder fairness objective.

Automated AI research. A small but growing body of work delegates the design of ML pipelines to LLM coding agents. Karpathy's autoresearch [4] runs a coding agent that modifies `train.py` for nanoGPT pretraining and is rewarded for validation loss. Our system shares this autoresearch architecture (coding agent + frozen evaluation harness + diff-based history) but targets a different inner loop (multi-agent policy synthesis instead of single-model pretraining) and a different objective (multi-agent social welfare instead of validation bits-per-byte). To our knowledge this is the first instantiation of the autoresearch paradigm in a multi-agent decision-making domain.

Automated mechanism and information design. Classical automated mechanism design [7] computes optimal allocation rules for self-interested agents under explicit incentive constraints, while information design [8] studies how a principal commits to a signaling policy that shapes a receiver's behavior. Our outer agent solves a related but distinct problem: $\mathcal{M}$ is not strategically deceptive (it follows instructions) but is *boundedly rational*, so the researcher must decide what information, helpers, and structure to expose so that $\mathcal{M}$ writes policies achieving the principal's welfare goal. Section 4 shows that the pipelines $\mathcal{R}$ produces are genuinely a function of the welfare objective, supporting this information-design framing empirically.

6 Discussion and Conclusion

We have presented a two-level framework where a coding agent autonomously discovers pipeline configurations that improve the output of an inner-loop LLM system. Applied to multi-agent policy synthesis in social dilemmas, the researcher agent reliably exceeds hand-designed baselines across multiple independent runs, and converges on qualitatively similar strategies within each game–objective condition. The framework requires no domain-specific scaffolding: the agent operates on a standard software repository using the same tools (file editing, shell commands, git) available to a human researcher.

Mechanism design in action. Our results empirically support the mechanism design interpretation of Section 3.4. The researcher acts as an information designer: under Φ_U, it reveals efficiency-oriented information (waste counts, zone assignments) that guides $\mathcal{M}$ toward productive but unequal role allocation (Listing 5). Under $\Phi_{\min}$, it additionally reveals fairness-oriented structure such as rotation schedules and per-agent equity feedback (Listings 3, 4), guiding $\mathcal{M}$ toward egalitarian coordination.

Two types of dilemma. The Cleanup–Gathering contrast reveals a structural insight about social dilemmas: *provision* dilemmas (asymmetric costs, one agent's contribution benefits all) require explicit fairness mechanisms, while *restraint* dilemmas (symmetric costs, each agent faces the same temptation) achieve fairness naturally when coordination improves. This distinction, well-known in the common-pool resource literature, is rediscovered here by an autonomous AI agent, which suggests it is a robust structural feature of these environments.

Human oversight and audit. Our system is fully autonomous by design, but its architecture offers natural affordances for human-in-the-loop oversight. Because the researcher $\mathcal{R}$ operates on a standard git repository,a domain expert can audit the full history of code diffs Δ_j alongside their metric outcomes $(\Phi_j, \mathbf{m}_j)$. The keep/discard decision (Algorithm 1, line 11) is a lightweight intervention point: a human could override the accept threshold, veto modifications that introduce undesirable mechanisms (e.g., exploitative role assignments), or steer the researcher toward under-explored regions of the configuration space by editing the experiment history. More broadly, the separation between the researcher $\mathcal{R}$ and the frozen evaluation Φ creates a *delegation boundary*: the human defines *what* to optimize (the welfare objective), while the agent decides *how*. This mirrors the expert-in-the-loop design patterns identified in deployment-oriented work on AI agents for discovery, where trust calibration depends on the legibility of the agent's decision trail rather than on real-time supervision.

Future work. Several directions emerge. First, we are interested in applying the two-level framework to other LLM-driven pipelines (code optimization, scientific experiment design, infrastructure tuning), where the same architecture applies with a different inner loop and objective Φ; next, adversarial objectives can be intriguing, to test whether the researcher discovers exploitative pipeline configurations, exposing reward hacking risks; and asymmetric programs: extend to settings where agents run *different* source code (the present setup is symmetric in code but already supports heterogeneous behavior via `agent_id`; relaxing the shared-program constraint enables richer per-agent specialization and partial-information settings).

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A Additional Results

B Compute Requirements

Table 4 reports wall-clock time for all 12 autonomous runs and the 6 GEPA comparison runs. *Inner loop* measures total time spent on policy LLM generation and environment simulation across all evaluations; *total* (estimated from run-directory timestamps) additionally includes the researcher agent's analysis, code editing, and decision-making between evaluations.

Table 3: Strategies discovered by the researcher in Cleanup across 4 efficiency-optimized and 4 maximin-optimized independent runs.

Strategy	Φ_U (4 runs)	$\Phi_{\min}$ (4 runs)
Waste-counting helpers	4/4	4/4
Zone/lane partitioning	3/4	4/4
Anti-regression feedback	3/4	2/4
Worked policy examples	2/4	3/4
Cleaning cost economics	2/4	3/4
Time-based role rotation	**0/4**	**4/4**

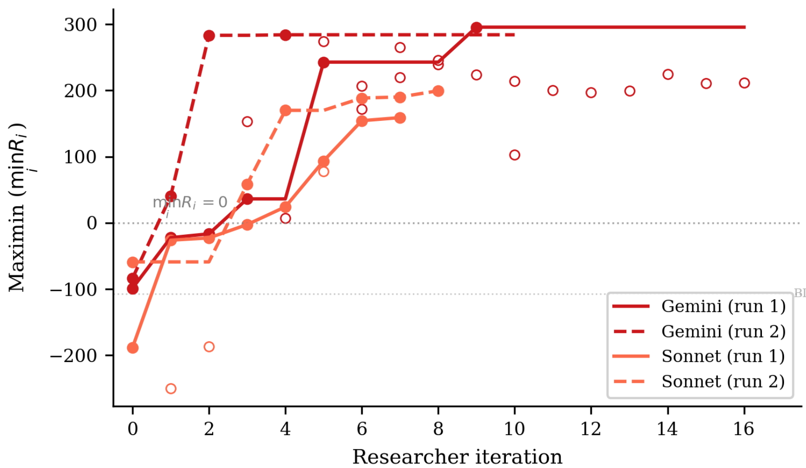

Figure 3: Maximin ($\min_i R_i$) across researcher iterations for the 4 Cleanup maximin-optimized runs. All runs transform deeply negative baselines (worst-off agents losing reward) into substantially positive values. Gemini reaches $\sim$290 while Sonnet reaches $\sim$160–200. The dashed line marks $\min_i R_i = 0$ (no agent loses reward overall).

Cost breakdown. Each inner loop evaluation involves K=2–3 policy LLM generation calls (each $\sim$2k input tokens, $\sim$1.5k output tokens) followed by multi-seed simulation (5–12 seeds, $\sim$15–30s total). Policy LLM generation dominates inner loop time (86–97%), with Sonnet evaluations taking $\sim 2\times$ longer than Gemini due to extended thinking. The researcher agent (Claude Opus 4.6, running via Claude Code CLI) accounts for 41% of total wall-clock time (25.6h of the 62.2h total across all 12 runs), spent reading results, editing pipeline source files, and planning modifications. In monetary terms, the researcher agent is the dominant cost: each outer iteration consumes $\sim$50–100k context tokens in the Opus session, whereas each inner loop evaluation uses only $\sim$5–10k policy LLM tokens. The 6 GEPA comparison runs required 5.0h of compute with no researcher agent, operating at lower cost per run but achieving substantially lower performance (Table 2).

C Smaller Open-Weight Model: Gemma 4 26B

We test the framework with Gemma 4 26B-A4B-IT (Google), a 26B-parameter open-weight model substantially smaller than the frontier models in the main experiments. We run three experiments—two on Cleanup (N=10) optimizing efficiency and maximin respectively, and one on Gathering (N=4) optimizing efficiency—all with Opus 4.6 as the researcher $\mathcal{R}$.

Setup. In all three experiments, Gemma starts from complete failure: the baseline pipeline produces U=−10.0 on Cleanup (agents CLEAN-spam without collecting apples) and U=0.0 on Gathering (broken BFS calls), compared to U=1.93/0.86 (Gemini/Sonnet on Cleanup) and U=2.04/0.03 (Gemini/Sonnet on Gathering). The researcher ran J=10–18 outer iterations per condition.

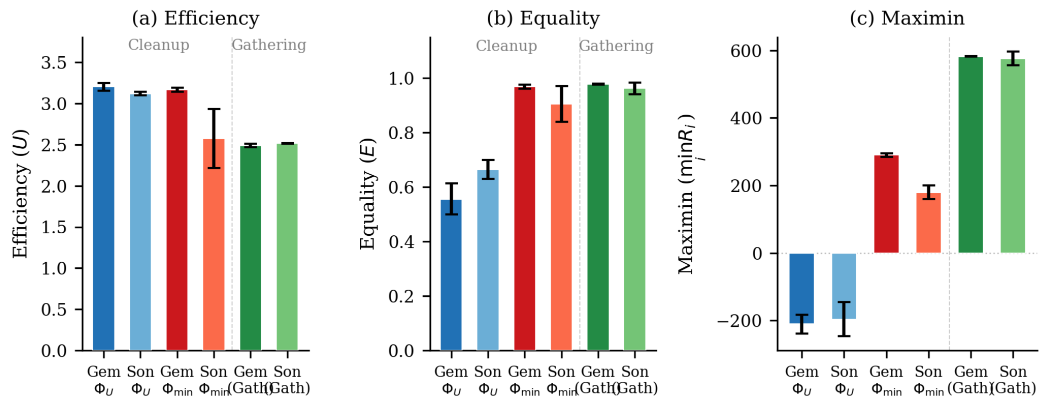

Figure 4: Final metrics across all conditions. (a) Efficiency: all conditions converge to $U \approx 2.5$–3.2. (b) Equality: maximin-optimized Cleanup runs achieve $E \approx 1.0$, while efficiency-optimized runs show $E \approx 0.5$–0.7; Gathering achieves high equality regardless. (c) Maximin: the sharpest contrast—efficiency optimization leaves the worst-off agent deeply negative ($\min_i R_i \approx -200$), while maximin optimization transforms it to $+290$ (Gemini) or $+179$ (Sonnet). Gathering achieves high maximin ($\sim$580) under efficiency optimization alone. Error bars show s.d. across runs.

Table 4: Wall-clock time per autonomous run. Evals: total inner loop evaluations including initial baseline. Per eval: mean wall-clock per single evaluation. Total: end-to-end including researcher overhead.

Policy LLM	Φ	Evals	Inner loop (h)	Per eval (min)	Total (h)
Cleanup (N=10)					
Gemini	Φ_U	11	2.6	14	3.7
Gemini	Φ_U	18	4.1	14	6.4
Sonnet	Φ_U	4	2.1	32	6.1
Sonnet	Φ_U	7	5.0	43	6.1
Gemini	$\Phi_{\min}$	17	3.0	11	4.3
Gemini	$\Phi_{\min}$	11	3.6	20	5.0
Sonnet	$\Phi_{\min}$	8	4.5	33	8.8
Sonnet	$\Phi_{\min}$	9	5.3	36	13.2
Gathering (N=4)					
Sonnet	Φ_U	3	1.7	33	2.2
Gemini	Φ_U	5	1.4	17	1.9
Gemini	Φ_U	3	0.9	19	1.1
Sonnet	Φ_U	4	2.3	35	3.4
All 12 runs		100	36.6	22	62.2
GEPA (6 runs)		6	5.0	50	—

Results. Table 5 presents the best configurations discovered. Under efficiency optimization, the researcher achieves U=0.87—a dramatic recovery from the broken baseline but far below frontier models (U≈3.2). The researcher compensates for the model's limited capability by reducing inner-loop iterations to K=1 (avoiding the regression that plagued K≥2 runs) and providing highly structured worked examples with explicit cleaning-role assignment.

Maximin optimization rescues efficiency. Strikingly, under maximin optimization the researcher achieves *higher* efficiency (U=1.71) than under direct efficiency optimization (U=0.87), alongside strong equality (E=0.94) and positive worst-off welfare ($\min_i R_i$=137). This reversal—absent in frontier models, where both objectives yield U≈3.1–3.2—occurs because the maximin objective forces the researcher toward coordination mechanisms (rotating cleaning duties, index-based apple assignment) that simultaneously improve collective output. Under efficiency-only optimization, the researcher converges on a local optimum—static role assignment with 25% dedicated cleaners—that a 26B model can implement but that caps performance well below the frontier.

Table 5: Autoresearch with Gemma 4 26B-A4B-IT. Single run per condition. Baseline: pipeline from [3].

Game	Target	U	E	$\min_i R_i$	Keep rate
Cleanup	Baseline	-10.0	$1.00^\dagger$	-1000	—
	Φ_U	0.87	-1.11	-371	6/19
	$\Phi_{\min}$	**1.71**	**0.94**	**137**	9/17
Gathering	Baseline	0.0	$1.00^\dagger$	0	—
	Φ_U	**2.44**	**0.98**	**580**	4/11

†Equality is trivially 1.0 because all agents receive near-zero reward.

This suggests that *for models below a capability threshold, fairness objectives may serve as better optimization targets for overall social welfare than direct efficiency maximization*. The structured coordination enforced by the maximin objective provides a scaffold that compensates for the weaker model's difficulty implementing complex strategies from strategic hints alone.

Gathering: full recovery on a simpler game. On Gathering (N=4), the researcher brings Gemma from U=0.0 to U=2.44 with E=0.98 and $\min_i R_i$=580, after 10 outer iterations (4 kept). These values nearly match frontier models (Gemini: U=2.49, Sonnet: U=2.52; Table 2). The researcher discovers the same Voronoi partitioning and respawn-aware camping strategies found in the main experiments, and increases inner-loop iterations to K=5 to allow sufficient refinement. The contrast with Cleanup—where Gemma peaks at U=0.87/1.71 versus frontier $U\approx$3.2—suggests that the researcher can fully compensate for model weakness on simpler coordination tasks (4 agents, symmetric costs) but not on harder provision dilemmas (10 agents, asymmetric costs).

D Researcher-Authored Pipeline Artifacts

The previous appendix shows the policies π_c^* that the inner loop *outputs*; this one shows the configuration $c = (p, \phi, \mathcal{H}, \iota)$ (Section 4, Table 1) that the researcher $\mathcal{R}$ *authored* to produce them. Excerpts are taken from the final commit on the dedicated git branch of the corresponding run; we reproduce the prose verbatim, with author commentary in [brackets] and elisions marked . . . . We organize by artifact type so each subsection makes a within-type contrast (e.g., Φ_U vs. $\Phi_{\min}$, or Cleanup vs. Gathering).

D.1 Helper library $\mathcal{H}$: coordination primitives

The researcher adds primitives that the policy LLM can call as black boxes, sparing it from re-implementing tricky logic on every iteration. Two patterns dominate: (i) *state inspection* (waste counts, fractions, beam-yield scoring) and (ii) *coordination primitives* (zone assignment, role rotation). We show one of each.

Listing 1: Cleanup, $\Phi_{\min}$ – band-based apple zoning helper added by the researcher in exp5. This is the primitive that lets the policy in Listing 6 keep collectors within their own row band, preventing all 7 gatherers from racing to the same apple.

```python
def get_my_apples(env, agent_id):
    """Alive apples in this agent's horizontal row band.
    Divides the orchard into n_agents bands; falls back to all apples if empty."""
    aid     = int(agent_id)
    n       = int(env.n_agents)
    band_h = env.height / n
    band_lo, band_hi = aid * band_h, (aid + 1) * band_h

    my_apples, all_apples = set(), set()
    for i in range(env.n_apples):
        if env.apple_alive[i]:
            ar = int(env._apple_pos[i, 0]); ac = int(env._apple_pos[i, 1])
            all_apples.add((ar, ac))
            if band_lo <= ar < band_hi:
                my_apples.add((ar, ac))
    return my_apples if my_apples else all_apples
```

Listing 2: Gathering – BFS-Voronoi territory + respawn-aware waiting helpers added by the researcher in gather-exp3. These are the primitives that Listing 8 composes into a one-line policy: "go to my_zone_apples, otherwise nearest_respawning_apple."

```python
def voronoi_zones(env):
    """Multi-source BFS Voronoi over walkable cells.
    Returns (row, col) -> agent_id; ties broken by lower agent_id."""
    queue, visited = deque(), {}
    for a_id in range(env.n_agents):
        if int(env.agent_timeout[a_id]) == 0:
            ar = int(env.agent_pos[a_id][0]); ac = int(env.agent_pos[a_id][1])
            queue.append((ar, ac, a_id))
            visited[(ar, ac)] = a_id
    while queue:
        r, c, a_id = queue.popleft()
        for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
            nr, nc = r + dr, c + dc
            if 0 <= nr < env.height and 0 <= nc < env.width \
                    and not env.walls[nr, nc] and (nr, nc) not in visited:
                visited[(nr, nc)] = a_id
                queue.append((nr, nc, a_id))
    return visited

def nearest_respawning_apple(env, agent_id, zones=None, max_timer=10):
    """Nearest dead apple in MY zone respawning within max_timer steps.
    Returns (row, col) of best wait spot, or None."""
    if zones is None: zones = voronoi_zones(env)
    best_pos, best_t, best_d = None, max_timer + 1, float('inf')
    ar = int(env.agent_pos[agent_id][0]); ac = int(env.agent_pos[agent_id][1])
    for i in range(env.n_apples):
        if not env.apple_alive[i]:
            pos = (int(env._apple_pos[i][0]), int(env._apple_pos[i][1]))
            if zones.get(pos) == agent_id:
                t = int(env.apple_timer[i])
                if t <= max_timer:
                    d = abs(pos[0] - ar) + abs(pos[1] - ac)
                    if t < best_t or (t == best_t and d < best_d):
                        best_pos, best_t, best_d = pos, t, d
    return best_pos
```

The choice of helper depends on the dilemma's geometry: row-bands suffice when fairness is the main concern (Cleanup maximin) but full BFS-Voronoi is needed when wall-aware territory ownership matters (Gathering). The researcher does not fix this in advance — it picks the simpler primitive whenever it works.

D.2 System prompt p: from neutral framing to strategic briefing

The unmodified system prompt is a neutral API description: *"Write a policy that maximizes per-agent reward."* Under maximin optimization, the researcher rewrites the prompt into a multi-section strategic briefing. The excerpt below shows the pieces it added to the Cleanup prompt in exp5; the full prompt grew from 165 to 325 lines.

Listing 3: Cleanup, $\Phi_{\min}$ – excerpts from the researcher-rewritten system prompt (exp5). The unmodified baseline contained only API documentation; everything below was added by $\mathcal{R}$.

```
## CRITICAL OBJECTIVE: Rawlsian Fairness (Maximin)

Your goal is to maximize the minimum per-agent total return across all
agents. This is the "maximin" or Rawlsian welfare criterion.

This means: it is NOT enough for the *average* agent to do well. The
worst-off agent must do as well as possible. A policy where 8 agents
each earn +200 but 2 agents each earn -100 is TERRIBLE (maximin = -100).

Key implication: cleaning costs (-1 per CLEAN action) must be shared
equitably among ALL agents. If you assign fixed "cleaner" roles, those
agents will accumulate large negative rewards and destroy your maximin score.

## Strategy for Maximin: ROLE ROTATION + APPLE ZONING

The optimal strategy for maximin involves:
1. Shared cleaning duty: ALL agents take turns cleaning using a rotation
   schedule based on `agent_id` and `env._step_count`. For example:
   `is_my_cleaning_turn = (agent_id + env._step_count // SHIFT) % env.n_agents < NUM_CLEANERS`
   where SHIFT is ~50 steps and NUM_CLEANERS is 2-3.
2. When NOT your turn: collect apples from YOUR ZONE using
```

```
   ‘get_my_apples(env, agent_id)‘ -- prevents all gatherers competing
   for the same nearest apple.
3. NEVER use BEAM (action 6): it costs -1 and causes -50 to the target.
4. Keep waste density low: aim for ~2-3 active cleaners at any time.

[... API documentation, helper documentation, working example ...]

IMPORTANT:
- NEVER assign permanent cleaning roles to specific agents -- this kills maximin.
- Use env._step_count for time-based role rotation so all agents share cleaning duty.
- NEVER use BEAM (action 6) -- it destroys both agents’ rewards.
```

Two things stand out. First, the researcher writes the formula it expects $\mathcal{M}$ to use almost verbatim — `(agent_id + env._step_count // SHIFT) % env.n_agents < NUM_CLEANERS` — and the policy in Listing 6 uses essentially this template. Second, the researcher *teaches by counter-example*: the explicit “8 agents earn +200, 2 earn -100, maximin = -100” makes the failure mode of Φ_U-style static roles (Listing 5) concretely visible to $\mathcal{M}$. This is the information-design move predicted by the mechanism-design framing in Section 3.4.

The Gathering prompt evolves along a different axis (no maximin, no rotation). Its researcher-added “Key Strategic Insights” section instead emphasizes *(i) self-play implies never beam* and *(ii) Voronoi partition each step*, exactly matching what the policy in Listing 8 implements.

D.3 Feedback ϕ: adaptive diagnostics

The baseline feedback function from [3] shows reward + four social metrics with definitions and stops there. The researcher turns it into a state machine: it inspects the latest metrics and conditionally injects different hints. Two different runs produced two qualitatively different diagnostics — the same metric ($\bar{r}_k$ or $\min_i R_i$) is repurposed to either *stabilize* a working policy or *redirect* a failing one.

Listing 4: Adaptive feedback diagnostics. *Top*: efficiency-optimized run (exp1) – a stability guard prevents iter-3 regression once U is high. *Bottom*: maximin-optimized run (exp5) – two fairness diagnostics that fire when the worst-off agent loses reward.

```python
# --- Cleanup, Phi_U feedback (exp1, pipeline/feedback.py final) ---
last_eff = history[-1]["metrics"]["efficiency"]
if last_eff >= 2.5:
    parts.append(
        "**CRITICAL -- DO NOT REGRESS**: The current policy achieves high "
        "efficiency (>=2.5). You MUST output a policy that is nearly identical "
        "to the current one. Copy the current policy code and make AT MOST "
        "one small targeted improvement (e.g., adjust a single numeric "
        "threshold). If you are not confident a change will help, output the "
        "current policy UNCHANGED. A regression here means the run fails.")

# --- Cleanup, Phi_min feedback (exp5, pipeline/feedback.py final) ---
last_maximin = history[-1]["metrics"].get("maximin", 0)
last_avg     = history[-1]["reward_avg"]
if last_maximin < 0:
    parts.append(
        f"**FAIRNESS ALERT**: maximin={last_maximin:.1f} is NEGATIVE. "
        f"The worst-off agent lost reward overall while average was {last_avg:.1f}. "
        "This means cleaning duties are NOT shared equitably. "
        "Use time-based role rotation (env._step_count) so ALL agents share "
        "cleaning costs equally. NEVER assign permanent cleaner roles.")
elif last_maximin < last_avg * 0.5:
    parts.append(
        f"**FAIRNESS WARNING**: maximin={last_maximin:.1f} is much lower than "
        f"average={last_avg:.1f}. The gap suggests unequal cleaning burden. "
        "Ensure ALL agents rotate through cleaning duty.")
```

The two diagnostics are written by independent researcher runs but converge on a common pattern: *thresholded interventions on a primary metric*. They are also a common cause of the low run-to-run variance reported in Finding 1: when a working policy lands in the high-U basin, the stability guard prevents the LLM from over-refining and tipping out of it (the $K{=}3$ regressions visible in Section 4’s “Common failure modes” (4)), and when an unfair policy lands in the negative-maximin region, the alert injects exactly the rotation hint that recovers it.

D.4 Iteration logic ι: per-condition hyperparameters

Although ι has the smallest source surface, the researcher's edits to it explain a meaningful fraction of the variance reduction noted in Finding 1. Table 6 summarizes the values shipped in the final commit of each best-performing run.

Table 6: Iteration-logic (ι) values in the final commit of selected runs. K = inner-loop iterations, $|S|$ = evaluation seeds, "thinking" = extended-thinking token budget passed to $\mathcal{M}$. Baseline ([3]) is K=3, $|S|$=5, thinking 16k.

Run	Game / objective	K	$\|S\|$	thinking	Researcher's stated rationale
exp1 (Gemini)	Cleanup, Φ_U	3	5	16k	default; stability comes from feedback hint
exp4 (Sonnet)	Cleanup, Φ_U	3	5	32k	"Sonnet was over-refining at 16k thinking"
exp5 (Gemini)	Cleanup, $\Phi_{\min}$	**2**	**12**	16k	"K=2 avoids iter-3 regression; 12 seeds for variance control"
exp7 (Sonnet)	Cleanup, $\Phi_{\min}$	3	5	**10k**	"reduced thinking from 16k – Sonnet was generating runaway code at 16k"
gather-exp3 (Gemini)	Gathering, Φ_U	3	5	16k	default; Gathering converges in $K \leq 3$

The maximin Gemini run is the most informative case: the researcher discovered that with K=3 and $|S|$=5, identical configurations could yield $\min_i R_i$=295 on one set of seeds and $\min_i R_i$=100 on another (this is the "variance exposure" failure mode of Section 4's common failures). It then walked $|S|$ from $5 \rightarrow 8 \rightarrow 12$ over three outer iterations until the seed-averaged signal was stable enough that the policy LLM stopped chasing noise. The policy in Listing 6 is sampled at $|S|$=12.

D.5 Cross-artifact observations

- *Helpers do the algebra; prompts pick the strategy class.* The researcher uses helpers (1, 2) to put non-trivial primitives at $\mathcal{M}$'s fingertips, and the prompt (3) to tell $\mathcal{M}$ *which* primitive to compose. Neither is sufficient alone — prompts without supporting helpers produced policies that "mention rotation" but failed to implement it; helpers without prompt updates often went unused.
- *Feedback is the only adaptive component.* p, $\mathcal{H}$, and ι are static within a run; ϕ is the only place where the researcher gets to change what $\mathcal{M}$ sees *during* the inner loop, and it uses thresholded diagnostics (Listing 4) to do so. This is what mostly drives the run-to-run variance reduction.
- *The same artifact has different content under each objective.* The Cleanup prompt under Φ_U omits "rotation" and "maximin" entirely; the same prompt under $\Phi_{\min}$ devotes its entire opening section to it. The configuration c is genuinely a function of the welfare objective Φ, not a fixed pipeline parameterized by it — consistent with the information-design framing of Section 3.4.
- *The researcher's edits are short and targeted.* The full diff between the baseline pipeline and the best maximin Gemini configuration totals ~300 added lines spread over four files; ~80 of them appear in the listings above. The remaining content is API documentation and worked examples, included for completeness but not strategically novel.

E Selected Generated Policies

The code excerpts below are verbatim outputs of the policy synthesizer $\mathcal{M}$, taken from the final inner-loop iteration of the best-performing run in each condition. To fit the page, we elide repeated boilerplate (BFS scaffolding, fallback branches) with `...` and add commentary in `[brackets]`; nothing else has been edited. The listings illustrate the qualitative differences highlighted in Findings 2–4 of Section 4: *static* vs. *rotating* cleaning roles in Cleanup, and the *Voronoi* + *respawn-timer* structure in Gathering. All four policies were independently generated by the LLM under different seeds, models, and outer-loop runs.

E.1 Cleanup, Φ_U (Gemini): static interleaved roles + dynamic threshold

Best run: U=3.25, E=0.61, $\min_i R_i = -182$. The synthesizer assigns *permanent* cleaning roles via `agent_id` $\bmod n < \tau$, where the threshold τ depends smoothly on the current waste fraction. Higher pollution recruits more cleaners, but the *identity* of each cleaner is fixed — agents 0–$\tau-1$ always pay the cost. This drives U to its ceiling but leaves equality at 0.61 and pushes the worst-off agent to $\min_i R_i$=$-$182.

Listing 5: Cleanup, Φ_U – Gemini 3.1 Pro. Static role assignment with waste-tiered threshold (verbatim, abridged).

```
def policy(env, agent_id) -> int:
    if int(env.agent_timeout[agent_id]) > 0:
        return 7  # STAND while removed

    waste_frac = waste_fraction(env)
    n_agents = env.n_agents

    # [Smooth dynamic threshold: more pollution -> more cleaners]
    if   waste_frac > 0.35: threshold_cleaners = n_agents
    elif waste_frac > 0.25: threshold_cleaners = int(n_agents * 0.7)
    elif waste_frac > 0.15: threshold_cleaners = int(n_agents * 0.5)
    elif waste_frac > 0.05: threshold_cleaners = int(n_agents * 0.3)
    else:                   threshold_cleaners = max(1, int(n_agents * 0.2))

    # [STATIC role: cleaners are ALWAYS the lowest-id agents -- no rotation]
    is_cleaner = (agent_id % n_agents) < threshold_cleaners

    if is_cleaner:
        best_orient, n_waste = find_best_clean_orientation(env, agent_id)
        if n_waste >= 1:
            cur_orient = int(env.agent_orient[agent_id])
            if best_orient == cur_orient: return 8  # CLEAN
            if (cur_orient + 1) % 4 == best_orient: return 5  # ROTATE_RIGHT
            elif (cur_orient - 1) % 4 == best_orient: return 4  # ROTATE_LEFT
            else: return 4
        # [Competitor-aware BFS toward waste in own row band]
        c_idx = agent_id % n_agents
        target_r = int(((c_idx + 0.5) / threshold_cleaners) * env.height)
        # ... pick waste cell minimizing dist + |row - target_r| + 50 *
            #closer_cleaners
        # ... bfs_toward and direction_to_action
    else:
        # GATHERER: row-banded sweep, competitor-aware
        # assigned_row = ((g_idx + 0.5) / n_gatherers) * env.height
        # ... pick alive apple minimizing dist + |row - assigned_row| + 50 *
            #closer_gatherers
        ...
```

E.2 Cleanup, $\Phi_{\min}$ (Gemini): time-rotated, geographically interleaved roles

Best run: U=3.19, E=0.98, $\min_i R_i$=296. Switching the outer objective from U to $\min_i R_i$ flips the role-assignment scheme entirely: `role_idx` now depends on `agent_id` + `step`/50, so every agent rotates through cleaner and gatherer duty in a 50-step cycle. The cleaner indices $\{1, 4, 8\}$ are spatially *interleaved* along the river (rather than contiguous) so that whoever is on duty has a short walk to their assigned river slice. This single change accounts for the E=0.55 $\rightarrow$ 0.98 jump and the $\min_i R_i$=$-182 \rightarrow 296$ swing reported in Table 2, with negligible efficiency loss.

Listing 6: Cleanup, $\Phi_{\min}$ – Gemini 3.1 Pro. Time-rotated roles with interleaved geography (verbatim, abridged).

```
def policy(env, agent_id) -> int:
    if int(env.agent_timeout[agent_id]) > 0:
        return 7  # STAND while removed

    aid  = int(agent_id)
    step = int(env._step_count)
    n    = int(env.n_agents)

    # [TIME ROTATION: every 50 steps each agent's role index advances]
    if n == 10:
        role_idx = (aid + step // 50) % 10
        # [Cleaner indices {1,4,8} are SPATIALLY INTERLEAVED, not contiguous,
        #  so each on-duty cleaner has a short walk to its river slice]
        if   role_idx == 1: role_type, zone_idx = 'C', 0
        elif role_idx == 4: role_type, zone_idx = 'C', 1
        elif role_idx == 8: role_type, zone_idx = 'C', 2
        else:
            role_type = 'G'
            g_map = {0:0, 2:1, 3:2, 5:3, 6:4, 7:5, 9:6}
            zone_idx = g_map[role_idx]
        num_cleaners, num_gatherers = 3, 7

    if role_type == 'C':
        # [Each on-duty cleaner owns one horizontal slice of the river]
```

```
        row_min = int( zone_idx      * env.height / num_cleaners)
        row_max = int((zone_idx + 1) * env.height / num_cleaners) - 1

        # [Score every (row, col, orientation) in the slice by beam yield;
        #  fire if facing >=2 waste, else move to a >=2-yield position]
        best_positions, good_positions = set(), set()
        for r in range(row_min, row_max + 1):
            for c in range(10):
                if env.walls[r, c] or env.waste[r, c]: continue
                y_max = max(get_clean_yield(env, r, c, o) for o in range(4))
                if y_max >= 2: best_positions.add((r, c))
                if y_max >= 1: good_positions.add((r, c))
        # ... rotate / CLEAN / bfs_to_target_set as appropriate
    else:
        # GATHERER: collect apples ONLY inside the rotating row band
        # row_min/row_max for num_gatherers slices ...
        ...
```

E.3 Cleanup, $\Phi_{\min}$ (Sonnet): independent rediscovery of duty rotation

Best run: U=2.93, E=0.83, $\min_i R_i$=154. Run with a different policy LLM (Sonnet 4.6) and on a different outer-loop trajectory, the synthesizer arrives at the *same* structural insight as the Gemini maximin run: a phase counter $(\texttt{agent\_id}+\texttt{step}/50) \bmod n$ rotates which 2 of the 10 agents clean at any given time, and a *separate* 200-step zone counter rotates which 5-row band each collector sweeps. The two rotation periods (50 and 200 steps) ensure every agent visits every cleaning slot *and* every apple zone within one episode — a structural fairness invariant the researcher discovered without any explicit instruction to "rotate."

Listing 7: Cleanup, $\Phi_{\min}$ – Sonnet 4.6. Phase-rotated cleaning + zone-rotated collection (verbatim, abridged).

```
def policy(env, agent_id) -> int:
    if int(env.agent_timeout[agent_id]) > 0:
        return 7  # STAND while removed

    n    = env.n_agents
    step = env._step_count
    wf   = waste_fraction(env)

    # --- ROLE ASSIGNMENT: fair 2-cleaner rotation ---
    # [50-step phases x 20 phases / episode -> each agent cleans 4 phases (~20%)]
    phase        = step // 50
    cleaner_rank = (agent_id + phase) % n    # [cycles 0..9 fairly]
    is_cleaner   = cleaner_rank < 2          # [exactly 2 cleaners per phase]

    # [Emergency override: all agents clean when waste crosses the spawn cliff]
    in_emergency = wf >= 0.40
    if in_emergency: is_cleaner = True

    # [River split: cleaner_slot 0 -> top half, 1 -> bottom half]
    cleaner_slot = (agent_id % 2) if in_emergency else cleaner_rank

    if is_cleaner and wf > 0.04:
        # [Scan 4 orientations from current position; if best beam shot
        #  covers >=3 waste cells, fire; else step to a strictly better neighbor]
        # ... (best_o, best_cnt) = max over 4 directions
        # ... if best_cnt >= 3: rotate to best_o, then CLEAN
        # ... else: scan 4 adjacent cells x 4 orientations for higher yield
        # ... if no waste in beam range: bfs to waste_set in OWN river half,
        #     fall back to whole river

    # --- COLLECTOR LOGIC: 5-zone rotating apple collection ---
    # [5 zones x 200 steps = 1000 steps -> each agent visits every zone once;
    #  2 agents share each zone at any time, reducing competition 10 -> 2]
    bounds     = zone_boundaries(env, 5)
    zone_phase = step // 200
    zone       = (agent_id + zone_phase) % 5
    z_start, z_end = bounds[zone], bounds[zone + 1]

    zone_apples = {(int(env._apple_pos[i][0]), int(env._apple_pos[i][1]))
                   for i in range(env.n_apples) if env.apple_alive[i]
                   and z_start <= int(env._apple_pos[i][0]) < z_end}
    # ... bfs_to_target_set(zone_apples), else bfs_nearest_apple as fallback
    return 7
```

E.4 Gathering (Gemini): wall-aware Voronoi + spatiotemporal targeting

Best run: U=2.47, E=0.98, $\min_i R_i$=580. In Gathering, where costs are symmetric, no role differentiation is needed: the entire optimization is spatial and temporal. The synthesizer (i) partitions cells into BFS-Voronoi territories owned by individual agents, (ii) runs a single centralized BFS from its own position to score every reachable cell with exact wall-aware distances, and (iii) for both alive and respawning apples in its territory, computes the *earliest collection time* $\max($`walk_dist`, `respawn_timer`$)$ and targets the minimum. When the agent must wait on a dead apple, it steps onto a *non-spawn* adjacent cell rather than blocking a respawn point. This single sort key — $\max(\text{distance}, \text{timer})$ — subsumes both "go to nearest apple" and "camp respawn" as special cases.

Listing 8: Gathering – Gemini 3.1 Pro. Voronoi territories + spatiotemporal priority targeting (verbatim, abridged).

```
def policy(env, agent_id) -> int:
    if int(env.agent_timeout[agent_id]) > 0:
        return 7

    r, c   = int(env.agent_pos[agent_id][0]), int(env.agent_pos[agent_id][1])
    orient = int(env.agent_orient[agent_id])

    # [Wall-aware Voronoi over the gridworld; spawn_points = all apple cells]
    zones        = voronoi_zones(env)
    spawn_points = {(int(p[0]), int(p[1])) for p in env._apple_pos}

    # [Single centralized BFS: exact distance + first move to every reachable cell]
    distances, first_moves = {}, {}
    q = deque([(r, c, 0, None)])
    visited = {(r, c)}
    while q:
        cr, cc, d, fm = q.popleft()
        distances[(cr, cc)]   = d
        first_moves[(cr, cc)] = fm
        for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
            nr, nc = cr + dr, cc + dc
            if 0 <= nr < env.height and 0 <= nc < env.width and not env.walls[nr,
                nc]:
                if (nr, nc) not in visited:
                    visited.add((nr, nc))
                    q.append((nr, nc, d + 1, fm if fm is not None else (dr, dc)))

    # [Both alive AND respawning apples in MY zone, with their respawn timer]
    my_zone_spawns = []
    for i in range(env.n_apples):
        pr, pc = int(env._apple_pos[i][0]), int(env._apple_pos[i][1])
        if zones.get((pr, pc)) == agent_id and (pr, pc) in distances:
            my_zone_spawns.append((pr, pc, int(env.apple_timer[i]), distances[(pr,
                pc)]))

    # [SPATIOTEMPORAL PRIORITY: earliest collection time first.
    #  score = max(walk_distance, respawn_timer); ties broken by sooner timer]
    my_zone_spawns.sort(key=lambda x: (max(x[3], x[2]), x[2], x[3]))

    for pr, pc, timer, dist in my_zone_spawns:
        if timer == 0:                    # [Alive apple: walk straight to it]
            if dist == 0: return 7
            fm = first_moves[(pr, pc)]
            return direction_to_action(fm[0], fm[1], orient)
        else:                             # [Dead apple: camp on safe adjacent cell]
            # ... pick nearest neighbor (nr,nc) that is NOT in spawn_points,
            #     navigate there and STAND while waiting for respawn
            ...
    # [Global poaching fallback if zone is empty: nearest alive apple anywhere]
    ...
```

E.5 Cross-condition observations

Several patterns are visible across these four listings (and confirmed by inspecting the remaining 8 runs not shown):

- *Role assignment encodes the welfare objective.* Static `agent_id < ` τ (Listing 5) maximizes U but harms $\min_i R_i$. Time-rotated `(agent_id + step//`T`) % n` (Listings 6, 7) maximizes $\min_i R_i$ at $\leq 1\%$ efficiency cost (Gemini).

- *Convergent rediscovery across LLMs.* Gemini and Sonnet, on independent runs, both arrive at the `(id + phase) % n` duty-rotation idiom with similar phase lengths ($T \approx 50$ steps) under maximin — and both omit it under efficiency.
- *Game structure dictates strategy class.* Cleanup policies are dominated by role-assignment logic (cleaner vs. gatherer); Gathering policies are dominated by territory partitioning and respawn-timer reasoning, with no role differentiation at all. The researcher does not need to be told which class of strategy to write.
- *Earliest-collection-time as a unifying score.* The Gathering policy's $\max(\text{walk}, \text{timer})$ key (Listing 8) is structurally identical across the 4 Gathering runs (both Gemini and Sonnet), differing only in how the Voronoi diagram is computed — Manhattan approximation, fully BFS-based, or cached helper.